Cranfield UNIVERSITY

ICECREMO II

Extensions of the Ice Growth and Water Film Models: Anti-icing and de-icing capabilities.

DELIVERABLE 3.1.2

by

C.P. Thompson

May 2004

Version 1.1

Commercial in Confidence

Applied Mathematics and Computing Group
School of Engineering
Cranfield University, Cranfield, Beds, MK43 0AL, UK

Contents

1 Introduction **4**

1.1 The Icecremo I Model 4

1.1.1 The flow problem 5

1.1.2 The thermal problem 6

2 Anti-Icing **7**

2.1 Water Layer 7

2.2 Conduction in the airframe 8

3 De-icing **9**

3.1 Heat transport in the 'Ice Layer' 9

4 Discussion **12**

4.1 Validation Plan 12

4.2 Implementation Plan 12

4.3 Effort 12

A Typical energy terms **13**

Disclaimer

Standard disclaimers apply.

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, without the prior permission in writing of the copyright owner.

Date	Version	Comments	Author
1.0	17/5/04	Original Draft	CPT
1.1	20/5/04	Boundary Conditions, Other post-meeting comments	CPT

Summary

This report extends the ice growth and water runback models of Icecremo I to deal with the cases of anti-icing and de-icing.

Three separate improvements are described:

- Extension to anti-icing systems: treatment of runback in the presence of strong heating.
- Modelling of heating elements and conduction in the airframe skin.
- A further enhancement to the model will allow the limited treatment of de-icing situations. This is physically complex and a simplified model is outlined.

The report ends with a discussion of the implementation and testing strategy.

Nomenclature

a_T, a_F — constants in the specification of the thermal boundary conditions applied at the bottom of the airframe/substrate
(A_1, A_2) — surface shear in the (x, y) directions
b — ice height
c_w — specific heat capacity of water
F_i — thermal flux boundary conditions at the bottom of the airframe surface.
(g_x, g_y, g_z) — acceleration due to gravity in the (x, y, z) directions
h — water height
k_i, k_s, k_w — thermal conductivity of ice, airframe/substrate, water,
L_e, L_f — latent heat of evaporation, latent heat of freezing
p — water pressure
p_a — external air pressure
q_0 — energy source terms
q_1 — energy sink terms
q_{0r} — energy source terms (for rime ice case)
q_{1r} — energy sink terms (for rime ice case)
$\mathbf{Q}$ — liquid flux
t — time
T — ice temperature
T — temperature in the generalised ice layer (de-icing model)
$\hat{T}$ — airframe temperature
T_f — freezing temperature
T_i — temperature boundary conditions at the bottom of the airframe surface.
T_s — temperature at airframe surface (Icecremo I model)
W — free stream velocity
x, y — lateral coordinate directions
z — distance/height normal to the airframe surface

Greek

β — catch efficiency
ϵ — thickness of the freezing zone approximation
κ_w — thermal diffusivity of water $\kappa_w = \frac{k_w}{\rho_w c_w}$
κ_s — thermal diffusivity of the substrate (airframe skin) $\kappa_s = \frac{k_s}{\rho_s c_s}$
μ_w — viscosity of water
ρ_A — droplet content of the air.
ρ_i — density of ice (898 kg m^{-3})
σ — surface tension
θ — temperature in the water layer

1 Introduction

A number of discussions have been held to consider the extension of the ice models to be incorporated into the Icecremo II code. The goal is to be able to support anti-icing and de-icing protection systems. The purpose of this note is to survey possible models which will support this aim and to summarise the alternatives to allow a clearer definition of the work plan. This will include a discussion of the limitations of each model and how these can be quantified.

For clarity we will start by describing the current model used in Icecremo I. Note that the focus of this document is on mathematical models, one of the key differences between Icecemo I and Icecremo II is that the former uses a structured grid on the surface of the airframe to calculate the ice growth and water runback. Icecremo II uses an unstructured grid ([1]).

1.1 The Icecremo I Model

Icecremo I is a predictive tool capable of computing the growth of both rime ice and glaze ice. In the latter case the movement of runback water is also calculated. A model is derived which starts from the full three-dimensional Navier-Stokes equations. However, these can be simplified under the observation that we are dealing with **thin** films, in which case the model falls into the general class of lubrication models which can be formulated in terms of liquid film height. Equally important, the ice height is determined by an energy balance which yields a Stefan type condition. In general these two methods yield a theoretically consistent and reasonably accurate model of the key physics of icing.

The project only considered fixed boundary conditions. The airframe thermal boundary conditions must be specified as either a fixed temperature or heat flux. Similarly the external conditions: air speed, temperature, droplet size et al. are also fixed. This simplified the model development significantly. See the paper by Myers, Charpin and Thompson for the derivations ([2]).

We set the scene here by re-stating the key equations. In subsequent sections we will discuss possible enhancements to this model. The first is to consider the anti-icing case that is the current scope of the Icecremo II proposal. In this case we consider heating areas of the airframe and hence we need to generalise the thermal model to support the surface temperature (T_s) and the effects of water significantly above freezing. The second extension is to include de-icing strategies. In this case heating elements can be switched on

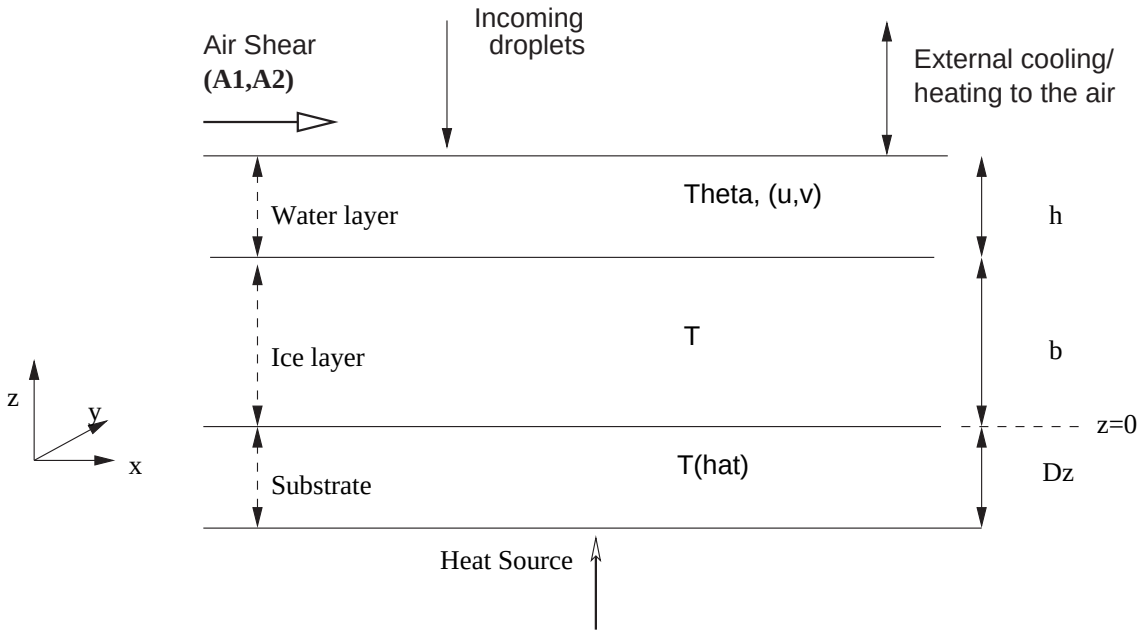


Figure 1: Basic geometry

and off and this allow the development of thermal profiles in the ice layer and the possibility of warm water layers being created under an ice layer. In the full generality the ice/water configurations can be complex with multiple layers of water trapped under domains of ice and sections of ice becoming freed because of large areas of water underneath. It is likely that a simpified model will be able to deal with all practically encountered situations and the scope of the model will be discussed later.

1.1.1 The flow problem

The Icecremo I model is derived from the Navier Stokes equations by using the fact that the aspect ratio of the flow (the depth scale:length scale) is small and this allows major simplifications of the differential system. The basic geometry and some of the key variable names are give in Figure (1). The lateral water velocity components (u, v) are given by:

$$\mu_w u = \frac{1}{2}\left(\frac{\partial p}{\partial x} - \rho_w g_x\right)(z^2 - b^2 - 2(z-b)(b+h)) + A_1(z-b) \qquad (1)$$

and

$$\mu_w v = \frac{1}{2}\left(\frac{\partial p}{\partial y} - \rho_w g_y\right)(z^2 - b^2 - 2(z-b)(b+h)) + A_2(z-b) \qquad (2)$$

The w component contains terms involving the incoming water and the movement of the freezing front:

$$w = \frac{z-b}{h}\left[\frac{\rho_i}{\rho_w}\frac{\partial b}{\partial t} + \frac{\partial h}{\partial t} + u\left(\frac{\partial b}{\partial x} + \frac{\partial h}{\partial x}\right) + v\left(\frac{\partial b}{\partial y} + \frac{\partial h}{\partial y}\right) - \frac{\rho_A}{\rho_w}\beta W\right] + \left(1 - \frac{\rho_i}{\rho_w}\frac{\partial b}{\partial t}\right) \qquad (3)$$

This expression is not required to solve the flow problem, but will be used in the convection equation (13) below. The pressure in the fluid satisfies:

$$p - p_a = \rho_w g_z (z-h) - \sigma\nabla^2(b+h) \qquad (4)$$

These can be combined with the continuity equation to give an equation for the film thickness:

$$\frac{\partial h}{\partial t} + \nabla.\mathbf{Q} = \frac{\rho_A}{\rho_w}\beta W - \frac{\rho_i}{\rho_w}\frac{\partial b}{\partial t} \qquad (5)$$

Where the fluid flux is given by:

$$\mathbf{Q} = \left(-\frac{h^3}{3\mu_w}\left(\frac{\partial p}{\partial x} + \rho_w g_x\right) + \frac{h^2}{2\mu_w}A_1, -\frac{h^3}{3\mu_w}\left(\frac{\partial p}{\partial y} + \rho_w g_y\right) + \frac{h^2}{2\mu_w}A_2\right) \qquad (6)$$

At the edge of a wetted surface we use the pre-cursor film approximation:

$$h = h_{prec}(= 10^{-6})\mathrm{m} \qquad (7)$$

At the edge of the computational domain we can use either: $h = 0$ or $\partial h/\partial n = 0$. However, both of these are unrealistic and the computation must be terminated when the liquid film crosses the computational boundary. Of course, the results are meaningful until this point.

This is a fourth-order, non-linear, degenerate partial differential equation which has been numerically validated in a number of applications. This is in addition to the analytical work reported in [2] and [3].

1.1.2 The thermal problem

Rime ice In this case, where the environment is so cold that there is no liquid water present then the ice height (b) is given by:

$$b = \frac{\rho_A}{\rho_i}\beta W t \qquad (8)$$

and the temperature profile in the ice (T) satisfies:

$$T = T_s + \frac{q_{0r} + q_{1r}T_s}{1 - q_{1r}b}z \tag{9}$$

Glaze ice If a water layer exists we can write the following temperature equations. In the ice layer we have:

$$T = T_s + (T_f - T_s)\frac{z}{b} \tag{10}$$

And in the water layer we have:

$$\theta = T_f + \frac{q_0 + q_1 T_f}{1 - q_1 h}(z - b) \tag{11}$$

Where h is the water film height. The thickness of the ice layer (b) is derived from the energy balance at the freezing interface and satisfies the Stefan condition:

$$\rho_i L_f \frac{\partial b}{\partial t} = k_i \frac{T_f - T_s}{b} + k_w \frac{q_0 + q_1 T_f}{1 - q_1 h} \tag{12}$$

The thermal problem is fully specified once the boundary conditions: q_0, q_1, q_{0r} and q_{1r} are specified. In brief, we include the energy sources in the q_0 and q_{0r} terms and the other terms represent the sink terms; details of these are given in the Appendix.

One of the achievements of this model is that it is able to predict correctly the thickness of the rime ice layer which is observed in glaze ice situations see Myers and Hammond ([4]).

2 Anti-Icing

Task 3.1.1 covers the development of a methodology for modelling anti-icing systems. The assumption is that the boundary conditions will remain constant but include heated areas of the airframe.

2.1 Water Layer

Icecremo I did not consider the use of ice protection methods. In the current project there is the possibility of strong heating sources and therefore we

have to model convection by run-back water. The convection equation in the water layer satisfies:

$$\frac{\partial \theta}{\partial t} + \frac{\partial u\theta}{\partial x} + \frac{\partial v\theta}{\partial y} + \frac{\partial w\theta}{\partial z} = \kappa_w \left(\frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} + \frac{\partial^2 \theta}{\partial z^2} \right) \tag{13}$$

This can be simplified using the equations for the velocity (1 and 2). This replaces Equation 11. At the ice-water interface we have the Stefan condition (Eqn 12); at the air-water interface we have:

$$\frac{\partial \theta}{\partial z} = q_0 + q_1 \theta \tag{14}$$

At the edge of a wetted surface we use a thermal boundary condition inherited from flow boundary conditions. The thinness of the pre-cursor film implies that it is in local thermal equilibrium with the adjacent material giving:

$$\begin{aligned} \theta(\underline{x}) = \theta_{prec}(\underline{x}) &= b(\underline{x}) \qquad \text{if } b(\underline{x}) > 0 \\ &= \hat{T(\underline{x})} \qquad \text{otherwise} \end{aligned}$$

The same restrictions about the limitations of the validity of these conditions as the film crosses the computational boundaries are inherited from the flow problem.

In anti-icing systems there is, of course, significant loss of water due to evaporation and we need to include this term in Equation (5). Rather than a full evaporation model we will simply evaluate the mass loss as simply:

$$\frac{\text{evaporative heat loss}}{L_e} \tag{15}$$

(It is also possible to use a correlation for the evaporative mass loss and derive the change in energy from this.)

Note that the effect of evaporation on the energy equation is already modelled in the q_1 term.

Following the approach taken in Icecremo I we will approximate the thermal variation in terms of the mean temperature in the water layer.

2.2 Conduction in the airframe

Under certain circumstances there can be significant conductive heat transfer in components of the airframe. A full thermal model is beyond the scope of

an icing code since there are interactions between the skin and any structural elements nearby. However, to be effective there must be provision to model this effect in some way as well as to specify thermal inputs from the heating elements.

This was considered at a WP 3.1 technical meeting ([5]) and it was proposed that a '2-1/2' D model of material be included. In the direction normal to the surface the domain would be the same thickness as the airframe skin. Lateral conduction can be included in the model, in addition to the main heat transfer direction normal to the surface.

$$\frac{\partial \hat{T}}{\partial t} = \kappa_s \left(\frac{\partial^2 \hat{T}}{\partial x^2} + \frac{\partial^2 \hat{T}}{\partial y^2} + \frac{\partial^2 \hat{T}}{\partial z^2} \right) \tag{16}$$

The simplest boundary conditions to associate with this equation are that $\hat{T(\underline{x})} = T_{amb}$ at the edge of the domain. This will be true sufficiently far away from heating elements; but as with the other equations this becomes a poor approximation if there is significant flux across the edge of the domain.

Numerical experimentation will be required to investigate how the number of elements normal to the surface affects the accuracy. However, it is assumed that a small number of cells in the vertical direction will be adequate to model the heat transfer in the z-direction.

The thermal input at the 'bottom' of this solid layer will be specified in user-supplied subroutines. A sample set of routines will be developed as part of this workpackage. Essentially this will allow the specification of the heat input in the form:

$$k_s \frac{\partial \hat{T}}{\partial z} = a_T T_i + a_F F_i \tag{17}$$

3 De-icing

3.1 Heat transport in the 'Ice Layer'

The main challenge to the mathematical modelling of ice formation introduced by de-icing strategies is that caused by the possibility of time dependent thermal boundary conditions. This is not intrinsically difficult. The numerical algorithms can easily be extended to have varying values of T_i and F_i. However, this allows much more complex behaviour in the solution. In particular one may encounter the situation where there is a water layer

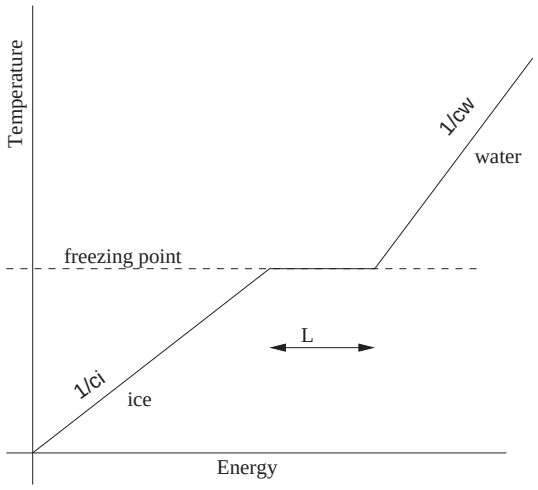


Figure 2: Temperature variation with Energy

adjacent to the air sitting over an ice layer which has formed before the ice protection system was activated. Beneath this ice layer could be a layer of water which has been created by the melting of the ice by the application of heat at the bottom of the domain. (In theory it could be possible to have a number of ice/water layers co-existing as the result of repeated switches of the ice protection system; however, this seems unlikely because of the resulting weakness of the ice layers and the possibility of shedding.)

This situation can be extremely difficult to model. Fortunately it is possible to make a number of simplifying assumptions. The first is to note that any liquid water trapped underneath ice is unlikely to have significant velocities. We can therefore ignore flow and convective effects and we only have to solve two equations; one continuity equation and one energy balance which is a conduction equation. The most accurate way of determining the location of the interface is to use a Stefan condition (Equation 12) at each water/ice interface. Even this approach is difficult because the appearance and disappearance of water layers is mathematically and computationally complex. An alternative approach, which avoids many of these difficulties was suggested by Dart [7].

Figure 2 shows the affect on the temperature of a volume of ice/water as the internal energy increase. It is clearly divided into three regions. In the central region there is no change in temperature: the energy is used to change the phase of the water. If we relax this condition then Figure 3 shows a different type of behaviour, although the graphs are quantitatively similar. Here we

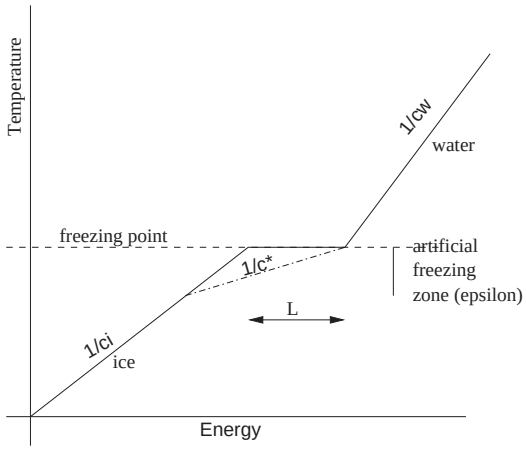


Figure 3: Temperature variation with Energy: showing 'freezing zone' approximation.

allow a slow increase in temperature as the internal energy increases in the interval $T_f - \epsilon < T < T_f$. The slope of the graph here is c_p^*. This is an artificial value introduced to approximate the absence of heat rise during the melting process. Under these assumptions we can write:

$$\rho^\phi c_p^\phi \frac{\partial T}{\partial t} = \frac{\partial}{\partial z} k^\phi \frac{\partial T}{\partial z} \tag{18}$$

where:

k^ϕ etc.	$=$	k_i	if $T < T_f - \epsilon$	ice
k^ϕ etc.	$=$	k^*	if $T_f - \epsilon < T < T_f$	mixed freezing zone
k^ϕ etc.	$=$	k_w	if $T > T_f$	water

Note that we have assumed that the conduction in the z direction dominates conduction in the lateral directions because the temperature differences are greater and the length scales are significantly less.

Simple geometric arguments show that if the freezing zone is defined by $(T_f - \epsilon, T_f)$ then we have:

$$c^\star = c_i + (L_f/\epsilon) \tag{19}$$

There is a difficult implementation issue to consider here. The implementation in Icecremo I uses semi-analytinc techniques to determine the temperature profile in the ice (this is also true of the current implementation

in Icecremo II). This means that the ice layer does not need to be discretised. Clearly this approach requires discretisation and this will increase the computational time (and storage requirements) for the code. This will be exacerbated by the difference between the conduction time-scales and those of the flow. (Of course, the dual time-step approach will help here.)

One issue which must be addressed is that, in principle, this approach supports various types of conditions which can lead to ice-shedding. This is beyond the scope of this workpackage and will be handled by producing appropriate error messages.

4 Discussion

4.1 Validation Plan

Individual software modules will be validated by solving special cases where analytic solutions can be obtained or special qualitative behaviour can be analysed. This is in addition to the implementation against experimental results which is in the Icecremo II proposal.

4.2 Implementation Plan

1. Software design — to support a full implementation of this model.
2. Substrate conduction will be implemented first.
3. Sample API to support different anti-icing and de-icing mechanisms.
4. Thermal convection by runback water.
5. Temperature variation in the 'generalised ice-layer'.

4.3 Effort

It is extremely difficult to estimate accurately the effort required to implement and validate this programme of work. It is a significant development over Icecremo I and, indeed, over the work programme specified in the Icecremo II proposal. We recommend that the individual capabilities are introduced into a version of the code which can be made available for testing

to consortium members. Progress should be reviewed at consortium meeting and priorities can be adjusted to match progress.

A Typical energy terms

At the air-water interface we typically have the following energy transfer mechanisms:

1. Latent heat: $Q_l = \rho_i L_f \frac{\partial b'}{\partial t'}$
2. Droplet kinetic energy: $Q_k = \frac{\rho_A \beta W^3}{2}$
3. Cooling from incoming droplets: $Q_d = \rho_A \beta W c_w (T'_A - T_f)$
4. Convective heat transfer: $Q_h = \overline{H}(T' - T_A)$
5. Aerodynamic heating: $Q_a = \frac{r\overline{H}W^2}{2c_a}$
6. Radiative heat flux: $Q_r = \epsilon\sigma_r(T'^4 - T_A^4) \approx 4\epsilon\sigma_r T_A^3(T' - T_A)$
7. Evaporation: $Q_e = \chi_e(e(T') - e(T_f)) \approx \chi_e e_0(T' - T_f)$ (If there is no water layer then we may have direct sublimation of the ice, which can be described in a similar fashion.

Further details can be found in [2] and the references cited therein.

References

[1] A.J. Press and D. Standingford *Icecremo II Software Design Methodology, JS15066* December 2003

[2] T.G. Myers, J.P.F. Charpin and C.P. Thompson *Slowly accreting ice due to supercooled water impacting on a cold surface* Physics of Fluids 2002

[3] H. C. W. Anderson *Stabilty Effects of Heat and Mass Transfer on Thin Liquid Films* 2002

[4] T.G. Myers and D. W. Hammond, *Ice and water film growth from incoming supercooled droplets.*Int. J. Heat and Mass Transf. **42** 2233 1999

[5] Nick Dart *Minutes of the 19/12/2003 Icecremo II Meeting* Dart et al.

[6] C.P. Thompson et al.(1998) *ICECREMO Software Design Specification Runback (Liquid Film) and Ice Formation Modules* ICECREMO/SPEC/CRAN2/CT980216/4.3

[7] Nick Dart (2003) *in [5]*

[8] D Warriner (2003) *Anti-ice protection user requirements Document* Deliverable 3.1.1.